

MAMA LINK

Multi Agent Maternal Health Coordination Solution

End to end solution design and production readiness plan

Final activity submission for the Microsoft Foundry hackathon

Prepared 18 September 2026



Submission position. MAMA-Link is a working synthetic hackathon prototype with deployed Foundry agents, deterministic safety tools, File Search grounding, Application Insights monitoring, and a tested local web workflow. It is not clinically validated and is not deployed for real patient care.

Executive summary

MAMA-Link addresses a coordination problem in maternal health: a person reporting danger signs may need risk recognition, appropriate facility capabilities, transport or social support, and a clear escalation path at the same time. The prototype coordinates these tasks through specialized agents while keeping safety-critical classification and referral selection under deterministic tools.

The implemented solution uses three Microsoft Foundry prompt agents: a Triage Agent, a Referral Agent, and a Knowledge Agent. The first two must call restricted tools and their final structured outputs are checked against authoritative local workflow results. The Knowledge Agent searches a small WHO and UNFPA public-source corpus through Foundry File Search. Application Insights receives only aggregate operation count, status, and duration. The local application provides intake, classification, capability-aware matching, history, consent-gated simulated referral, and an evaluation dashboard.

The evidence supports a hackathon demonstration: 25 automated tests pass; all 7 expected emergency fixtures are detected; 19 of 24 fixture classifications match; and all 19 expected nonroutine cases receive matching referral capabilities. Five expected-normal cases remain unassessed because no clinically approved low-risk completion protocol exists. The next production steps are independent clinical review, held-out cloud evaluation, authentication, live data agreements, and controlled deployment.

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1 Business scenario and intended users

1.1 Business problem

Maternal-health escalation often spans several disconnected decisions. A symptom report must be assessed consistently, an appropriate level of urgency must be communicated, a facility must be selected for required capabilities rather than proximity alone, and available support must be identified. Delays or unsupported assumptions at any step can prevent a person from reaching suitable care.

MAMA-Link demonstrates how an AI-assisted coordination workflow can organize these steps while preserving explicit human consent and deterministic safety controls. The system does not diagnose, prescribe, declare a person safe, or send a real alert.

1.2 Intended users

User	Need	Prototype support
Pregnant or postpartum person and family	Understand escalation status and available care options	Structured intake, plain status, educational links, consent-controlled simulated referral
Community health worker	Recognize risk and coordinate appropriate referral	Rule-based assessment, capability matching, traceable results
Facility or referral coordinator	Receive consistent referral context	Structured referral preparation; no live transmission in prototype
NGO or transport partner	Match assistance to service and coverage	Fictional support registry and availability filtering
Clinical and operational owner	Review safety, quality and system performance	Versioned rules, tests, evaluation report and privacy-safe monitoring

1.3 Business outcome

The target outcome is a shorter and more reliable path from a reported concern to an appropriate, consented care-coordination action. For the hackathon, success means demonstrating observable agent collaboration, grounded retrieval, correct tool use, repeatable evaluation, and clear safety limits.

2 Multi agent solution design

2.1 Specialized agents

Agent	Responsibility	Boundary	Implemented evidence
Triage Agent	Request the deterministic screening result for one synthetic case	Cannot change case ID, diagnose, prescribe, downgrade classification, or answer before tool use	mama-link-triage v1; live smoke test returned critical for MAT-005
Referral Agent	Request risk and capability-aware facility matches	Cannot invent a facility, mutate consent, dispatch, or reorder validated facility IDs	mama-link-referral v1; live smoke test returned FAC-009, FAC-002 and FAC-004

Agent	Responsibility	Boundary	Implemented evidence
Knowledge Agent	Retrieve educational content from the indexed corpus	Must use File Search, cite the source URL, acknowledge missing content, and avoid patient-specific advice	mama-link-knowledge v1; live UNFPA retrieval verified

2.2 Why multiple agents

The roles have different authority, tools, data, and failure modes. Separating them reduces the scope of each prompt and makes tool permissions reviewable. A deterministic triage output cannot be silently replaced by a fluent model answer; facility selection can be checked independently; and educational retrieval can be updated without changing emergency rules. Each role can also be evaluated, versioned, monitored, and approved separately.

A single general agent would hold broader permissions and mix classification, retrieval, facility selection, and communication in one response. That design would make failures harder to locate and increase the risk of unsupported clinical statements or invented resources.

2.3 Architecture

Web intake	Local API	Triage	Knowledge	Referral	User result
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The web application sends validated structured inputs to FastAPI. The authoritative local workflow applies versioned screening rules, derives required facility capabilities, matches fictional facilities and support organizations, and prepares a referral. Foundry agents demonstrate bounded orchestration over this workflow: the Triage and Referral Agents call read-only functions; the Knowledge Agent queries its vector store. User consent is required before the local application changes a referral from awaiting consent to simulated. No outbound notification integration exists.

3 Tools data and knowledge

Component	Used by	Purpose	Control
screen_case function	Triage Agent	Return deterministic classification for one requested fixture	Strict schema; exact case ID; read-only
referral_plan function	Referral Agent	Return classification and ordered capable facilities	Strict schema; read-only; validated final output
Foundry File Search	Knowledge Agent	Retrieve WHO and UNFPA educational material	One pinned vector store; pending clinical review label
Synthetic maternal dataset	Local workflow	24 fictional scenarios for regression and demonstration	No real patient data
Facility and support registries	Local workflow	Match capabilities, service area and simulated availability	Fictional records; no nearest-care claim
SQLite session store	Web application	Save assessment history and consent state	Browser-session isolation; raw notes excluded

3.1 Knowledge provenance

The prototype corpus records source URL, retrieval date and review status for WHO maternal mortality material, a WHO maternal and newborn counselling handbook, and the UNFPA obstetric fistula page. The corpus is intentionally labelled pending clinical review. It supports a retrieval demonstration and must not be described as an approved clinical protocol until a qualified owner records approval, jurisdiction, version, permitted use, and review date.

4 End to end workflow

4.1 Sequence of interactions

1. The user selects a synthetic fixture or enters structured fictional symptoms and measurements.
2. FastAPI validates the request. Free-text notes are treated as untrusted human-review context and are not interpreted or persisted.
3. The deterministic screening engine applies critical-over-warning precedence and returns classification, risk level, matched rules and required capabilities.
4. The Triage Agent calls `screen_case` and may return only the validated classification.
5. The Knowledge Agent searches its vector store for relevant educational material and returns source-linked information with the pending-review statement.
6. The Referral Agent calls `referral_plan`. The local workflow filters facilities by state, simulated availability and every required capability, then matches support organizations by service and coverage.
7. The application presents urgency, capable facilities, support options, evidence boundaries and a consent control.
8. If the user consents to a selected facility, the local record changes to simulated status. Nothing is sent externally.
9. The system records aggregate operation status and duration and exposes evaluation results to the dashboard.

4.2 Information passed between agents

Agents exchange the minimum structured context required for their role: a synthetic case ID for triage and referral, deterministic classification and facility IDs in validated outputs, and an educational question for retrieval. Raw session cookies, free-text notes, telemetry secrets and consent mutation authority are never passed to Foundry agents.

4.3 Failure behavior

- An agent response without its required tool call is rejected.
- A tool request for another case or another role's tool is rejected.
- A classification or facility list that differs from the authoritative workflow is rejected.
- The tool loop stops after five iterations.
- Unknown clinical input remains unassessed; it is never converted to a safety claim.
- Missing capable facilities produce an empty result rather than a fabricated fallback.

5 Observability strategy

5.1 Implemented monitoring

Application Insights is connected to a Log Analytics workspace with 30-day retention. The application emits an allow-listed counter and duration histogram using only operation name, status and elapsed milliseconds. Automatic FastAPI, requests, URL, Azure SDK and database instrumentation is disabled. This prevents raw symptoms, prompts, notes, case IDs, session IDs, request paths, tool payloads and secrets from entering telemetry.

Signal	Purpose	Diagnostic use
Operation count by status	Measure successful, rejected and failed operations	Detect regressions, rejection spikes and availability problems
Operation duration	Measure latency distribution	Find slow releases and establish performance targets
Agent tool-call status	Show whether a required tool completed	Identify orchestration or permission failures
Classification and referral evaluation metrics	Measure fixture behavior outside production telemetry	Detect rule regressions without logging health content

5.2 Production observability plan

A production dashboard would add availability, p50/p95 latency, agent and tool error rates, bounded-loop failures, retrieval success, source-citation presence, consent outcomes, referral acknowledgement and deployment version. Health-content metrics should be computed in a separately governed analytics process using approved de-identification and minimum aggregation thresholds. Alerts would cover sustained failures, latency breaches, missing grounding, abnormal tool rejection rates and telemetry gaps.

6 Evaluation strategy

6.1 Current test assets

Asset	Scope	Current result	Limit
Automated tests	Rules, API security, session isolation, consent, facilities and agent boundaries	25 of 25 pass	Regression tests over known implementation
24-case fixture evaluation	Classification and capability matching	19/24 labels; 7/7 emergencies; 19/19 nonroutine referral capability matches	Five normal fixtures remain unassessed
Foundry agent smoke test	Required tool use and exact output validation	Triage and referral passed for MAT-005	One known synthetic case
File Search smoke test	Grounded retrieval and source URL	UNFPA query passed	Small review-pending corpus

6.2 Evaluation criteria

Criterion	Measure	Proposed release gate
Task adherence	Required tool called; schema exact; no role expansion	100 percent on safety-critical suite
Tool usage	Correct tool and case; bounded completion	100 percent valid calls
Groundedness	Claims supported by retrieved approved source	No unsupported clinical claims

Criterion	Measure	Proposed release gate
Relevance and coherence	Response answers the request clearly	Human-reviewed threshold on held-out set
Safety	No diagnosis, prescription, reassurance, fabricated facility or false dispatch claim	Zero critical violations
Referral accuracy	All required capabilities present	100 percent on emergency cases
Emergency recall	Expected emergencies escalated	100 percent on approved benchmark
Latency and reliability	p95 completion and successful run rate	Target set after load test

6.3 Development lifecycle integration

Every rule, prompt, model, tool schema, corpus or facility-vocabulary change should create a versioned evaluation run. Pull-request checks run deterministic tests and the fixture evaluation. A staging gate runs held-out Foundry tests for grounding, safety, prompt injection, tool misuse and unavailable resources. Clinical and operational owners review failures and approve release evidence. After deployment, aggregate monitoring and sampled, authorized quality review feed new regression cases. Agent versions remain immutable so changes can be compared and rolled back.

7 Governance reliability and safety

Risk	Implemented control	Production requirement
Unsafe medical generation	Deterministic rules; prompts prohibit diagnosis/prescription; exact output validation	Clinical validation, content policy, red-team evidence and escalation ownership
False reassurance	No-match becomes unassessed, never low or safe	Clinically approved low-risk protocol
Fabricated facility	Capability and availability filter over fictional registry	Verified live directory, freshness SLA and fallback operations
Unauthorized referral	Explicit facility consent; simulated status; no dispatch tool	Strong identity, auditable consent and authorized partner integration
Sensitive-data leakage	Synthetic fixtures; notes not persisted; minimal telemetry	DPIA, retention policy, encryption, RBAC, private networking and incident response
Prompt injection	Case content treated as data; strict tools and schemas	Adversarial suite, content isolation and continuous monitoring
Model or service failure	Local deterministic workflow remains authoritative	Retries, timeouts, circuit breaker, operational fallback and rollback

7.1 Consistency and maintainability

Rules have a declared version and explicit precedence. Agent versions and vector-store identifiers are recorded in local ignored manifests. Infrastructure scripts are idempotent and validate existing Azure resources. Dependencies are bounded and an exact environment lock is maintained. Tests cover boundary conditions and cross-session access. Documentation records assumptions, known gaps, commands and acceptance evidence.

7.2 Human accountability

A production service requires named clinical, product, privacy, security and operations owners. Clinical owners approve screening and knowledge versions; operational owners verify facility and partner availability; privacy and security owners approve data flows; and release owners sign the evaluation record. The system supports decisions but does not replace professional judgement or emergency services.

8 Deployment and improvement lifecycle

Stage	Action	Evidence or gate
Develop	Change code, prompt, tool, rule or corpus in version control	Peer review and traceable change record
Verify	Run 25 tests and 24-case evaluation	No regression in emergency detection or safety boundaries
Evaluate	Run held-out Foundry quality, grounding, safety and adversarial suite	Approved release report
Deploy	Publish authenticated FastAPI service and frontend through controlled Azure environment	Managed identity, secrets, TLS, RBAC and health checks
Monitor	Observe aggregate reliability, latency, tool failures and safety indicators	Dashboard, alert ownership and incident runbook
Improve	Convert authorized findings into tests and versioned changes	Agent comparison, approval and rollback path

8.1 Current deployment position

The Azure Foundry account, project, gpt-4.1-mini deployment, three agent roles, vector store, Application Insights component and Log Analytics workspace are provisioned in Sweden Central. The application itself remains loopback-only because production authentication and partner data agreements are not implemented. Judges can run the application locally and inspect the cloud resources with the signed-in Azure account.

9 Demonstration plan

9.1 Recommended six minute recording

Time	Screen	Narration and proof
0:00–0:40	Title and problem	Explain the fragmented maternal referral problem and intended users.
0:40–1:30	Architecture	Show the three agent responsibilities, restricted tools, File Search and deterministic authority.
1:30–2:40	Local web app	Open MAT-005, run assessment, show critical status, capable facilities and support matching.

Time	Screen	Narration and proof
2:40–3:20	Consent	Choose a facility and demonstrate simulated referral. State clearly that nothing is sent.
3:20–4:20	Foundry	Show the three agent versions, tool use and grounded UNFPA retrieval with source URL.
4:20–5:10	Evaluation	Show 25 passing tests, 7/7 emergency detection and explain the five unassessed normal fixtures.
5:10–5:40	Monitoring	Show Application Insights and the privacy-safe metrics design.
5:40–6:00	Reflection	Explain how specialization, tools, evaluation and monitoring moved the idea toward production readiness.

9.2 Suggested demonstration scenario

Use synthetic case MAT-005, based on the PRD's prolonged-labour scenario. The expected demonstration path is critical classification, emergency-obstetric capability matching, support matching, explicit facility consent and a simulated referral state. In Foundry, run the two-agent demo and the knowledge demo separately. This accurately reflects the current implementation: the local web workflow is complete, while the cloud knowledge agent has not yet been inserted into the web request path.

9.3 Recording checklist

- Use fictional information only and close terminals containing secrets.
- Start the local server and verify `/api/health` before recording.
- Keep the Foundry project, agent list, vector store, Application Insights and evaluation output ready in separate tabs.
- Show tool calls or structured outputs, not only the final user interface.
- State the prototype limits and pending clinical review without minimizing them.
- End with the production plan and the specific evidence required for release.

10 Current evidence and remaining work

Hackathon requirement	Status	Evidence or next step
Business problem and users	Complete	Defined in this report and PRD
At least two specialized agents	Complete	Three deployed Foundry agent roles
Tools and knowledge	Complete for prototype	Two strict functions and File Search vector store
Information flow	Complete	Implemented local workflow and bounded Foundry demonstrations
Observability	Complete for prototype	Application Insights and privacy-safe custom metrics

Hackathon requirement	Status	Evidence or next step
Evaluation strategy	Complete; cloud suite pending	Tests and fixture metrics exist; add held-out Foundry evaluation
Governance and reliability	Plan complete	Controls implemented; clinical approval and production governance pending
Deployable workflow	Partial	Local app and Azure agents available; authenticated public hosting pending
Recorded presentation	Ready to record	Six-minute script and checklist included

The submission demonstrates the required design and production-readiness thinking with working prototype evidence. It should not claim production or clinical readiness. The most valuable next engineering milestone is a held-out Foundry evaluation suite covering tool adherence, grounding, unsafe advice, injection attempts and missing-resource behavior, followed by authenticated staging deployment.

Appendix A Azure resources

Resource	Configured value
Resource group	foundry-hackathon-rg-0c39e178
Foundry account	mamalink-ai-258f106f5903
Foundry project	mama-link
Region	Sweden Central
Model deployment	gpt-4.1-mini, version 2025-04-14, Global Standard, capacity 10
Prompt agents	mama-link-triage v1; mama-link-referral v1; mama-link-knowledge v1
Vector store	mama-link-public-sources
Application Insights	mamalink-insights-258f106f5903
Log Analytics	mamalink-logs-258f106f5903; 30-day retention

Appendix B Reproduction commands

```

.\scripts\setup.ps1
.\scripts\start.ps1
.\venv\Scripts\python.exe scripts\verify.py
.\infra\check-foundry.ps1
.\venv\Scripts\python.exe -m mama_link.foundry doctor
.\venv\Scripts\python.exe -m mama_link.foundry demo --case MAT-005
.\venv\Scripts\python.exe -m mama_link.foundry knowledge-demo
.\infra\provision-monitoring.ps1

```

Appendix C References

- Microsoft. File search tool for Microsoft Foundry agents. <https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/tools/file-search>
- Microsoft. Function calling with Foundry agents. <https://learn.microsoft.com/en-us/azure/foundry/agents/how-to/tools/function-calling>
- Microsoft. Enable OpenTelemetry in Application Insights. <https://learn.microsoft.com/en-us/azure/azure-monitor/app/opentelemetry-enable>
- World Health Organization. Maternal mortality. <https://www.who.int/news-room/fact-sheets/detail/maternal-mortality>
- World Health Organization. Counselling for Maternal and Newborn Health Care. https://iris.who.int/bitstream/handle/10665/44016/9789241547628_eng.pdf
- United Nations Population Fund. Obstetric fistula. <https://www.unfpa.org/obstetric-fistula>
- MAMA-Link product requirements document and synthetic evaluation assets in the project repository.